

- Document Control
- Document Title: Business Requirements Document (BRD)
- System Name: Application
- Version: 1.0
- Prepared By:
- Date:
- Status: Draft

- Executive Summary
 - This BRD outlines the key business requirements for the Application as described in the provided requirements. The scope includes ingestion, transformation, and processing of customer and order data into curated tables using delta live tables.

- Business Objectives
 - Enable efficient data ingestion from csv files to delta tables for customer and order.
 - Ensure clean, duplicate-free, and validated data for core reporting tables.
 - Maintain historical data using SCD Type 2 for comprehensive analytics.
 - Provide aggregate spend insights by customer and date.

- In-Scope
 - Reading source csv data from defined volumes and loading into delta tables.
 - Schema enforcement for customer and order tables.
 - Data cleansing, including removal of nulls and duplicates.
 - Calculation of total amount in orders.
 - SCD Type 2 implementation for ordersummary.
 - Aggregation logic for customer spending.
 - Use of delta live tables for orchestration.

- Out-of-Scope
 - Real-time data ingestion from sources other than defined csv files.
 - External reporting interfaces or dashboards.
 - Data sources/schema modifications not listed.
 - Downstream integrations not described.

- Stakeholders
 - Business Users:
 - Data Engineering Team:
 - Data Governance Team:
 - Project Sponsor:

- Business Requirements

BR-APP-001 — Ingest Customer Data from Source CSV

- Description: Read customer data from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata and load to delta table.
- Business Value: Establishes foundational customer data for processing.
- Priority: High

BR-APP-002 — Ingest Order Data from Source CSV

- Description: Read order data from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata and load to delta table.
- Business Value: Enables order analytics and reporting.
- Priority: High

BR-APP-003 — Enforce Customer and Order Table Schema

- Description: Schema for customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId).
- Business Value: Ensures consistent data structure for downstream processes.
- Priority: High

BR-APP-004 — Derive TotalAmount Column in Order Table

- Description: Add column "TotalAmount" in order table ($\text{PricePerUnit} * \text{Qty}$).
- Business Value: Enables order value calculations.
- Priority: High

BR-APP-005 — Data Cleansing

- Description: Remove null and duplicate records from customer and order tables.
- Business Value: Improves data quality and reliability.
- Priority: High

BR-APP-006 — Create Ordersummary Table with SCD Type 2

- Description: Create and maintain ordersummary table (SCD Type 2); join customer and order on CustId; maintain history and handle changes.
- Business Value: Provides historical data and robust audit trail.
- Priority: High

BR-APP-007 — Aggregate Customer Spend

- Description: Create customeraggregatespend table; aggregate TotalAmount by Name and Date.
- Business Value: Supports spend analysis and reporting.
- Priority: Medium

BR-APP-008 — Delta Live Tables Implementation

- Description: Implement all business logic using delta live tables.
- Business Value: Ensures pipeline automation and monitoring.
- Priority: High
- Assumptions & Constraints
 - Source CSV files are available and correctly formatted.
 - Delta Lake environment is configured and accessible.
 - No support for unlisted attributes or schemas.
 - Processing limited to defined data sources.
- Success Metrics
 - 100% of ingested records conform to schema rules.
 - Zero duplicate or null records in curated tables post-processing.
 - Historical changes reflected accurately in ordersummary.
 - Aggregated spend table updated on each pipeline run.
- Risks & Mitigations
 - Risk: Source files missing or corrupted.
 - Mitigation: Add upstream file validation and monitoring.
 - Risk: Schema evolution impacts pipeline.
 - Mitigation: Periodic review and schema enforcement automation.
 - Risk: Performance bottlenecks in delta live tables.
 - Mitigation: Monitor and optimize transformations; scale resources as needed.
- Approval
 - Business Owner:
 - Data Engineering Lead:
 - Compliance/Governance: